

Summer 1978

Problem 1 (Su78). For each of the following either give an example or else prove that no such example is possible.

1. A non-abelian group.
2. A finite abelian group that is not cyclic.
3. An infinite group with a subgroup of index 5.
4. Two finite groups that have the same order but are not isomorphic.
5. A group G with a subgroup H that is *not* normal.
6. A non-abelian group with no normal subgroups except the whole group and the unit element.
7. A group G with a normal subgroup H such that the factor group G/H is *not* isomorphic to any subgroup of G .
8. A group G with a subgroup H which has index 2 but is *not* normal.

Problem 2 (Su78, Fa87, Sp93). Let R be the set of 2×2 matrices of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

where a, b are elements of a given field $\mathbb{F}$. Show that with the usual matrix operations, R is a commutative ring with identity. For which of the following fields $\mathbb{F}$ is R a field: $\mathbb{F} = \mathbb{Q}, \mathbb{C}, \mathbb{Z}_5, \mathbb{Z}_7$?

Problem 3 (Su78). Let A be a $n \times n$ real matrix.

1. If the sum of each column element of A is 1 prove that there is a nonzero column vector x such that $Ax = x$.
2. Suppose that $n = 2$ and all entries in A are positive. Prove there is a nonzero column vector y and a number $\lambda > 0$ such that $Ay = \lambda y$.

Problem 4 (Fa77, Su78). 1. Using only the axioms for a field $\mathbb{F}$, prove that a system of m homogeneous linear equations in n unknowns with $m < n$ and coefficients in $\mathbb{F}$ has a nonzero solution.

2. Use Part 1 to show that if V is a vector space over $\mathbb{F}$ which is spanned by a finite number of elements, then every maximal linearly independent subset of V has the same number of elements.

Problem 5 (Su78). Evaluate $\int_0^{2\pi} e^{(e^{i\theta} - i\theta)} d\theta$.

Problem 6 (Su78). Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be an entire function and let $a > 0$ and $b > 0$ be constants.

1. If $|f(z)| \leq a\sqrt{|z|} + b$ for all z , prove that f is a constant.
2. What can one prove about f if $|f(z)| \leq a|z|^{5/2} + b$ for all z ?

Problem 7 (Su78). Let $\{S_\alpha\}$ be a family of connected subsets of $\mathbb{R}^2$ all containing the origin. Prove that $\bigcup_\alpha S_\alpha$ is connected.

Problem 8 (Su78, Sp03). Let X and Y be nonempty subsets of $\mathbb{R}^n$. Define

$$d(X, Y) = \inf\{d(x, y) \mid x \in X, y \in Y\}$$

1. Suppose X contains only one point x , and Y is closed. Prove $d(X, Y) = d(x, y)$ for some $y \in Y$.
2. Suppose X is compact and Y is closed. Prove $d(X, Y) = d(x, y)$ for some $x \in X, y \in Y$.
3. Show by example that the conclusion of Part 2 can be false if X and Y are closed but not compact.

Problem 9 (Su78). Let $U \subset \mathbb{R}^n$ be a convex open set and $f : U \rightarrow \mathbb{R}^n$ a differentiable function whose partial derivatives are uniformly bounded but not necessarily continuous. Prove that f has a unique continuous extension to the closure of U .

Problem 10 (Su78). Suppose the power series

$$\sum_{n=0}^{\infty} a_n z^n$$

converges for $|z| < R$ where z and the a_n are complex numbers. If $b_n \in \mathbb{C}$ are such that $|b_n| < n^2 |a_n|$ for all n , prove that

$$\sum_{n=0}^{\infty} b_n z^n$$

converges for $|z| < R$.

- Problem 11 (Su78).** 1. Suppose f is analytic on a connected open set $U \subset \mathbb{C}$ and f takes only real values. Prove that f is constant.
2. Suppose $W \subset \mathbb{C}$ is open, g is analytic on W , and $g'(z) \neq 0$ for all $z \in W$. Show that

$$\{\Re g(z) + \Im g(z) \mid z \in W\} \subset \mathbb{R}$$

is an open subset of $\mathbb{R}$.

Problem 12 (Su78). Let $p_1, \dots, p_n$ be elements of $\mathbb{F}[x]$, where $\mathbb{F}$ is a field. Prove that the greatest common divisor of $p_1, \dots, p_n$ is 1 if and only if there is an $n \times n$ matrix over $\mathbb{F}[x]$ of determinant 1 whose first row is $(p_1, \dots, p_n)$.

Problem 13 (Su78). Let G be a finite multiplicative group of 2×2 integer matrices.

1. Let $A \in G$. What can one prove about
 - (i) $\det A$?
 - (ii) the eigenvalues of A ?
 - (iii) the Jordan or Rational Canonical Form of A ?
 - (iv) the order of A ?
2. Find all such groups up to isomorphism.

Note: See also ??.

Problem 14 (Su78). Let V be a finite dimensional vector space over an algebraically closed field. A linear operator $T : V \rightarrow V$ is called *semisimple* if whenever a linear subspace $E \subset V$ is invariant under T , that is $T(E) \subset E$, there is a linear subspace $F \subset V$ which is invariant under T and such that $V = E \oplus F$. Prove that T is semisimple if and only if V has a basis of eigenvectors.

Problem 15 (Su78). Let E be the set of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ which are solutions to the differential equation $f''' + f'' - 2f = 0$.

1. Prove that E is a vector space and find its dimension.
2. Let $E_0 \subset E$ be the subspace of solutions g such that $\lim_{t \rightarrow \infty} g(t) = 0$. Find $g \in E_0$ such that $g(0) = 0$ and $g'(0) = 2$.

Problem 16 (Su78). Let N be a *norm* on the vector space $\mathbb{R}^n$; that is, $N : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies

- $N(x) \geq 0$ and $N(x) = 0$ only if $x = 0$
- $N(x + y) \leq N(x) + N(y)$
- $N(\lambda x) = |\lambda|N(x)$

for all $x, y \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$. Prove that

1. N is bounded on the unit sphere.
2. N is continuous.
3. There exist constants $A > 0$ and $B > 0$, such that for all $x \in \mathbb{R}^n$, $A\|x\| \leq N(x) \leq B\|x\|$.

Problem 17 (Su78). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Suppose that $\mathbb{R}$ contains a countably infinite subset S such that

$$\int_p^q f(x) dx = 0$$

if p and q are not in S . Prove that f is identically 0.

Problem 18 (Su78, Fa99). Let $M_{n \times n}$ denote the vector space of real $n \times n$ matrices. Define a map $f : M_{n \times n} \rightarrow M_{n \times n}$ by $f(X) = X^2$. Find the derivative of f .

Problem 19 (Su78). Solve the differential equation $g' = 2g$, $g(0) = a$, where a is a real constant.

Problem 20 (Su78, Fa89). Suppose $f : [0, 1] \rightarrow \mathbb{R}$ is continuous with $f(0) = 0$, and for $0 < x < 1$, f is differentiable and $0 \leq f'(x) \leq 2f(x)$. Prove that f is identically 0.

Problem 21 (Su78). Prove that a linear operator $T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is diagonalizable if for all $\lambda \in \mathbb{C}$, $\ker(T - \lambda I)^n = \ker(T - \lambda I)$, where I is the $n \times n$ identity matrix.

Problem 22 (Su78). Prove that a linear operator $T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is diagonalizable if it commutes with its conjugate transpose T^* , i.e., $(T^*)_{ij} = \overline{T_{ji}}$.